Sharad Vanth

DevOps Engineer

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Portfolio: GitHub.io

TECHNICAL SKILLS

- **DevOps Technologies**: Jenkins, Ant, Docker, Kubernetes, CI / CD, Microservices, Terraform, Helm, Ansible, YAML, Apache Tomcat, Bash, Shell Scripting.
- **Security**: SonarQube, Owasp Dependency Check, Trivy
- **Programming Languages:** Python, C#.
- **OS**: Linux, CentOS 6, Ubuntu
- **Databases:** MySQL, PostgreSQL , Mongodb, Redis
- **Containerization & Orchestration Tools:** Docker, Kubernetes, AKS.
- **Cloud Services:** Amazon Web Services, Microsoft Azure.
- **Version Control:** Git, GitHub.

EDUCATION

University of Cincinnati, Master of Science, Information Technology | Cincinnati, OH **Aug 2023**

Relevant Courses(Information Technology, Human Computer Interaction, Principles of Cybersecurity, Cloud Computing, Tech for Mobileapps, Data-driven Insights, System Administration.

G. Pulla Reddy Engineering College, Computer Science and Engineering | Andhra Pradesh, India **July 2021**

Relevant Courses(OS, Data Structures, Computer Networks, Python, Cloud Computing, DBMS)

WORK EXPERIENCE

Company: Fifth Third **Sep 2023 – Till Date**

Role: DevOps Engineer

- Implemented Azure Services such as Entra ID (AAD), Azure VM, Azure Virtual Networks, Azure Resource Manager, SQL Database, Azure Service Principal, Azure App Service, Azure Blob Storage, Azure Container Registry, Azure Kubernetes Service.
- Developed automation scripting using Ansible-playbooks to deploy and manage Java applications across Linux servers. Using Ansible deployed web applications to K8's. Involved in writing the ansible playbooks for installing docker in multiple nodes.
- Implemented a CI/CD pipeline with Docker, Jenkins, and GitHub by virtualizing the servers for the Dev and Test environments with Docker and achieving needs by configuring automation with Containerization.
- Assisted in troubleshooting and resolving deployment issues, honing skills in identifying and mitigating challenges to improve overall system stability.
- Collaborated with other DevOps engineers to maintain a deployment and management strategy - Assisting with creating and maintaining an automation and monitoring framework.
- Utilized SonarQube for static code analysis and continuous inspection of code quality, implementing customized quality gates and automated code reviews to enhance software reliability.
- Played a key role in the projects by deploying the web applications, gaining exposure to cloud-based technologies, and contributing to the successful transition of applications to modernized environments.

Company: Cognizant **Jan 2021 – July 2022**

Role: DevOps Engineer

- Leveraged AWS cloud services, including CloudFormation templates, EKS, EC2, RDS, S3, IAM, and VPC, to guarantee the availability, security, and peak performance of AWS cloud-hosted applications.
- I recently configured automated backups for our application data by creating versioned AWS S3 buckets. This ensures data durability and seamless integration with our CI/CD pipeline for efficient development workflows.
- I've used Terraform to automate the deployment of AWS infrastructure, defining resources like EC2 instances, VPCs, and security groups as code.
- Implemented and maintained build processes using Apache Ant, effectively streamlining the software development lifecycle and enhancing build automation for project deployments.
- Leveraging Terraform, we automated the scaling, ensured reproducibility, and facilitated rapid deployment of our microservices architecture.

- Utilizing Ansible, automated deployment processes guarantee consistent and reliable application rollouts across environments, establishing a robust deployment framework with Ansible.
- Implemented Docker for microservices deployment, efficient CI/CD integration in AWS Cloud environment. Docker's lightweight containers enhanced scalability and deployment reliability.
- Implemented automated Kubernetes deployments using Helm charts, Orchestrated Docker containers in Kubernetes for a high-traffic web-application enhancing scalability and minimizing manual intervention.
- Utilized Jenkins pipelines to automate the deployment of microservices, improving deployment consistency and reliability.
- I've actively enhanced productivity by modifying existing CI/CD pipelines in Jenkins. These modifications included optimizing build processes, integrating additional testing stages.
- Proficient in utilizing Git for version control, including branching, merging, and resolving conflicts, to streamline collaborative software development processes.
- Enhanced application monitoring and logging via CloudWatch to boost real-time insight into system performance, complemented by Prometheus and Grafana dashboards for effective visualization.
- Implemented blue-green deployment strategies, ensuring zero downtime during application updates.

Environment : AWS, Maven, Docker, Jenkins, Prometheus, Kubernetes, Helm charts, Terraform, Ansible, AWS

Company: Sunshine Techno Systems **Aug 2019 – Aug 2020**
Role: Software Engineer

- Collaborated with the team for Mini – Project of developing an e-commerce portal using dotnet to create a responsive web pages using C#,ASP.net, Entity Framework, Docker Desktop, Kubernetes.
- As a team member uploaded the microservices to Azure (Cloud Concepts) through Docker Desktop, Kubernetes.
- Troubleshooting issues involved during the build and deployment of any application in SDP.
- I have hands-on experience with Front End HTML, CSS, and Cloud Concepts as a Full Stack developer in the Dot Net Track.
- Responsible to deploy the Docker images to the Azure Container.
- Learned and applied fundamental Python concepts (Pandas, NumPy, Matplotlib) to contribute to simple coding tasks, gradually building a foundation in programming logic.
- Assisted in identifying and fixing basic bugs in Python scripts, gaining exposure to debugging techniques and error resolution.
- Explored introductory Python libraries to perform basic data manipulation tasks, acquiring a basic understanding of data processing in Python.
- Contributed to a small-scale Python project, working under guidance to understand basic software development practices and project structures.

Environment: HTML, CSS, Azure, Docker Desktop, Azure Containers, C#, Microservices, Python.

PROJECTS

- Multi-Tier Cloud-based WebApp Deployment Automation
  - Automated the entire workflow using Jenkins CI/CD, integrating Maven and JDK for processing the Java-based project. Setting up infrastructure using Terraform, and configured nodes using Ansible to install Docker and Kubernetes.
  - Compiled, running tests and Packaging the Code using Maven. Utilized SonarQube for code quality checks and OWASP Dependency-Check for vulnerability scans.
  - Deployed artifacts to JFrog, built Docker images using Docker, tagged images, and scanned them for vulnerabilities using Trivy, saving results in HTML format for clear log explanation.
  - Deployed Docker images to Docker Hub Registry and deployed the web application to Kubernetes in a dedicated namespace. Created a service account and generated a token for Jenkins authentication, using a service load balancer to enable public internet access to the web application.
  - Monitored the web application and system-level metrics using Prometheus, Grafana, and Blackbox Exporter.

- Multibranch_E-commerce-MicroserviceApplication
  - Designed and implemented a robust CI/CD pipeline using Jenkins for a sophisticated multi-branch microservices web application, streamlining development workflows and ensuring seamless integration and deployment processes.
  - Leveraged Docker and Dockerfiles to meticulously craft and optimize containerized images, subsequently tagging and pushing them to DockerHub repositories. Orchestrated their deployment onto managed Kubernetes clusters via Azure Kubernetes Service (AKS), ensuring scalability, reliability, and efficient resource utilization.
  - Orchestrated infrastructure provisioning using Terraform, automating the setup of essential components. Employed Ansible to configure and install crucial tools like Jenkins, kubectl, and Java across nodes, ensuring consistency and reliability in the deployment environment.
  - Integrated GitHub webhooks to trigger automated workflows, facilitating rapid iterations and ensuring synchronization between development and deployment phases.

CERTIFICATIONS

- Microsoft Azure Fundamentals
- Microsoft Azure Administrator Associate
- AWS Cloud Quest

E-Learning Platform Certifications:

- Achieved certification in Python Data-Structures, Retrieving, Processing on Coursera
- Achieved certifications in DevOps, CI/CD, Jenkins through LinkedinLearnings
- Completed Certifications in Git, Linux, Docker, Kubernetes, DevOps, Terraform, Ansible on KodeKloud